ACT SUBJECT TEST & DIGITAL SAT MATH COURSES

Rabaa Farrag

01120486569

23 YEARS OF EXPERIENCE

Reach a 700+ in ACT, Digital SAT & EST Math

✓ FIRST SESSION IS FREE!

Don't watch others hit 700+ while you wait.

ACT2 • ACT SUBJECT TEST MATH LEVEL 1 • PRACTICE EXAM

ACT2 Exam 2

Full Practice Test • Test #2

Linear systems, similarity, probability, functions & logic. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

50	MCQ	60 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /50

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

ACT, Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

ACT SUBJECT TEST • MATH LEVEL 1 • TEST #2

Practice Questions

50 QUESTIONS

1 The tables of ordered pairs represent some points on the graph of line q and v. Which system of equations is represented by lines q and v?

Line q

X	-9	-3	2
Y	0	18	33

Line v

X	-4	0	10
Y	10	8	3

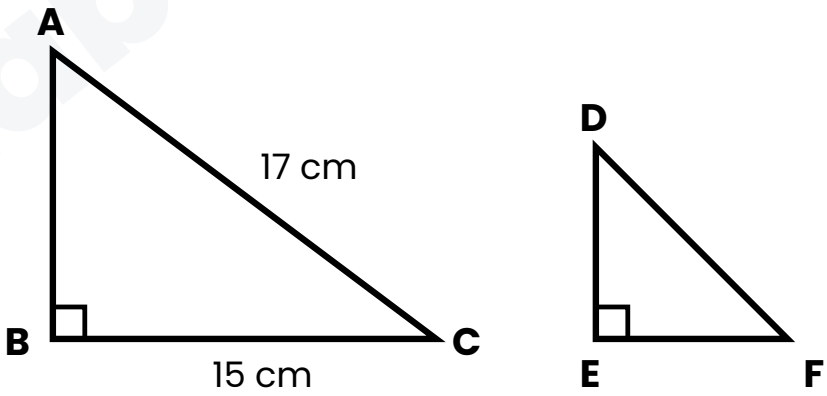
(A) $\begin{cases} 21x - y = 9 \\ 5x + 6y = 40 \end{cases}$

(B) $\begin{cases} 3x - y = -27 \\ x + 2y = 16 \end{cases}$

(C) $\begin{cases} 21x - y = 9 \\ 5x + 6y = 20 \end{cases}$

(D) $\begin{cases} 9x - y = -27 \\ x + 2y = 8 \end{cases}$

2 Kayle was cutting right triangles from wood to one of an art project. Two of the right triangles she cut are shown below. If $\Delta ABC \sim \Delta DEF$ with right angle B and E, $BC = 15$ cm and $AC = 17$ cm. What is the measure of $\angle F$ to the nearest degree?



(A) $28°$

(B) $41°$

(C) $62°$

(D) $88°$

Rabaa Farrag

3 Which equation represents the line passing through the points $(3, 2)$ and $(-9, 6)$?

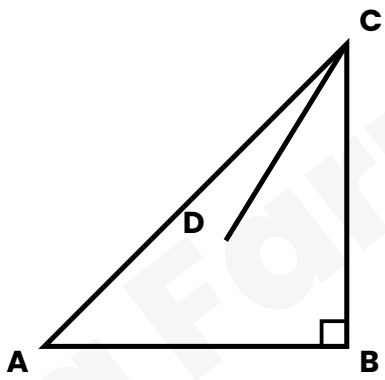
A $x - 3y = 9$
B $x + 3y = 9$
C $3x - y = -9$
D $3x + y = 9$

4 Determine the number of solutions based on the graph for the following system of equations:

$$\begin{cases} 2x + 5y = 7 \\ 10y = -4x + 14 \end{cases}$$

A Exactly one solution
B No solutions
C Infinitely many solutions
D Exactly two solutions

5 In the diagram below of ΔABC, $\angle ABC$ is a right angle, $AC = 12$, $AD = 8$ and altitude $\overline{BD}$ is drawn. What is the length of $\overline{BC}$?



A $4\sqrt{2}$
B $4\sqrt{3}$
C $4\sqrt{5}$
D $4\sqrt{6}$

6 For what value(s) of x is $\frac{2x+3}{5x+3} = \frac{2}{5}$?

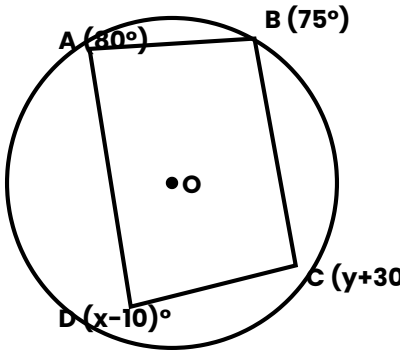
A $x = 1$
B $x = 3$
C No real numbers
D All real numbers

7 A circle whose center is the origin passes through the point $(-5, 12)$. Which point also lies on the circle?

A $(10, 3)$
B $(-12, 13)$
C $(11, 2\sqrt{12})$
D $(-8, 5\sqrt{21})$

Rabaa Farrag

8 Quadrilateral $ABCD$ is inscribed in circle O, as shown below. If $m\angle A = 80°$, $m\angle B = 75°$, $m\angle C = (y + 30)°$, and $m\angle D = (x - 10)°$, which statement is true?

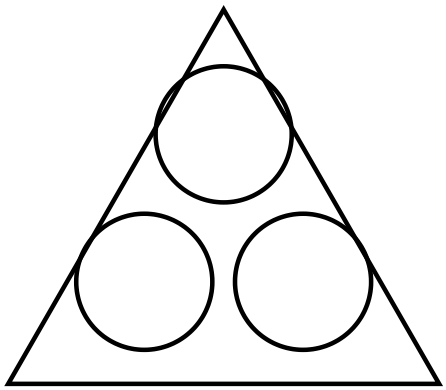


(A) $x = 85$ and $y = 50$
(B) $x = 90$ and $y = 45$
(C) $x = 110$ and $y = 75$
(D) $x = 115$ and $y = 70$

9 Which line divides any triangle into two triangles of equal area?

(A) A median
(B) An altitude
(C) An angle bisector
(D) A perpendicular bisector of a side

10 In the figure each of the three circles is tangent to the other 2. Each side of the triangle is tangent to two of the circles. Given that each circle has a radius of 4, what is the perimeter of the triangle?



(A) $8 + 8\sqrt{3}$
(B) $8 + 8\sqrt{2}$
(C) $24 + 24\sqrt{3}$
(D) $24 + 24\sqrt{2}$

Rabaa Farrag

11 A manufacturing company is building a rectangular room in their warehouse to store their products. The length of the room is 1 more than 3 times its width. The area of the room is 80 square meters. What are the dimensions of the room?

A $w = 4$ m, $l = 20$ m

B $w = 5$ m, $l = 16$ m

C $w = 5\frac{1}{3}$ m, $l = 15$ m

D $w = 8$ m, $l = 10$ m

12 A bag contains 16 red checkers and 16 black checkers. Anne removes 1 black checker and 2 red checkers from the bag and does not replace them. She then reaches into the bag and selects a checker at random. What is the probability that the checker is red?

A $\frac{1}{2}$

B $\frac{14}{15}$

C $\frac{14}{29}$

D $\frac{15}{29}$

13 Jasmine and her sister are saving to buy MP3 players. Jasmine has $50 and plans to save $10 per week. Her sister has $80 and plans to save $7 per week. In how many weeks will Jasmine have more money saved than her sister?

A 2 weeks

B 4 weeks

C 10 weeks

D 11 weeks

14 Thomas is a car salesman. The table shows the monthly salary that Thomas earns for the number of cars he sells. Use the data to write the equation of the line, identify the slope and explain what it means.

Number of cars sold	Monthly salary Earned
0	$2000
1	$2300
2	$2600
3	$2900
4	$3200

A $y = 2000x + 300$; slope is 300.

B $y = 300x + 2000$; slope is 300. This means that for every car Thomas sells, he earns an additional $300

C $y = 2000x + 300$; slope is 2000.

D $y = 300x + 2000$; slope is 2000.

Rabaa Farrag

15 If x is a positive odd integer and y is a negative odd integer such that $|x| \neq |y|$, which of the following must produce a positive even integer?

(A) y^x
(B) x^y
(C) $x - y$
(D) $\frac{y^2}{x}$

16 One side of an equilateral triangle measures 6 inches. What is the height of the triangle?

(A) $\sqrt{3}$
(B) $3\sqrt{3}$
(C) 6
(D) $6\sqrt{3}$

17 The trajectory of a potato launched from a potato cannon on the ground at an angle of $45°$ with an initial speed of 65 m/s can be modeled by the parabola $f(x) = x - 0.0023x^2$ where the x-axis is the ground. Find the height of the highest point of the trajectory and the horizontal distance the potato travels before hitting the ground.

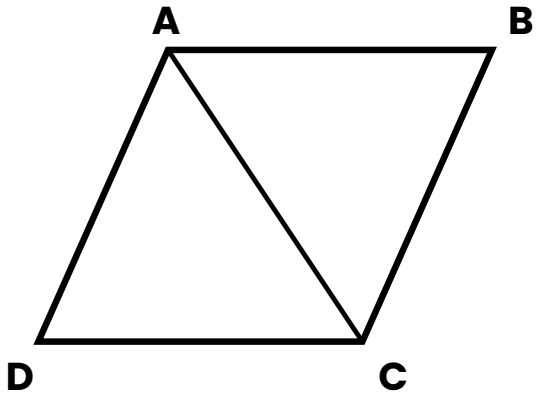


(A) height: 109 m; distance: 435 m
(B) height: 121 m; distance: 418 m
(C) height: 118 m; distance: 421 m
(D) height: 102 m; distance: 409 m

Rabaa Farrag

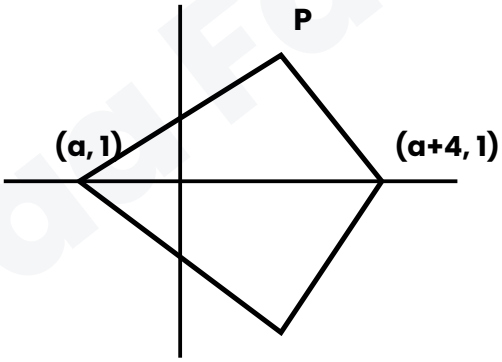
18 $ABCD$ is a rhombus. How do you complete the explanation that states why $\Delta ABC \cong \Delta CDA$?

Statement: $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$ by definition of rhombus. $\overline{AC} \cong \overline{AC}$ by Reflexive Property, so $\Delta ABC \cong \Delta CDA$ by...



A ASA
B AAS
C SAS
D SSS

19 The figure shown is a kite. What is the x-coordinate of point P?



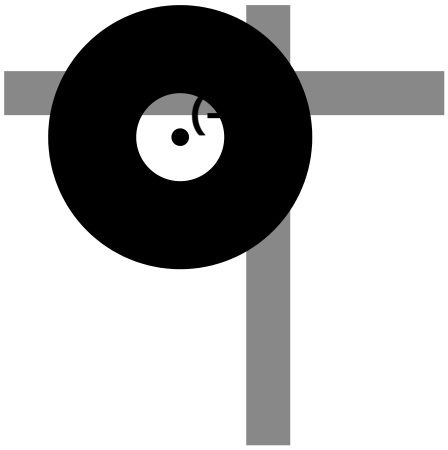
A $\frac{a}{2} + 4$
B $\frac{a+4}{2}$
C $a + 2$
D $2a + 4$

20 The solution of this system of inequalities forms a triangular region: $\begin{cases} y \leq 3x \\ y \leq -\frac{1}{2}x + 4 \\ y \geq 0 \end{cases}$. What is the area, in square units, of the region?

A $2\frac{2}{7}$
B $4\frac{4}{7}$
C $13\frac{5}{7}$
D $27\frac{3}{7}$

Rabaa Farrag

21 A manufacturing is designing a two-wheeled cart that can maneuver through tight spaces. On one test model, the wheel placement (Center) and radius is modeled by the equation $(x+2)^2+(y-1)^2=4$. What is the graph that shows the position and radius of the wheels?



A Circle with center $(-2,1)$ and radius 2
B Circle with center $(2,-1)$ and radius 2
C Circle with center $(-2,-1)$ and radius 4
D Circle with center $(2,1)$ and radius 4

22 What is the length of the diagonal of a rectangle whose perimeter is 34 and whose area is 60?

A 69
B 49
C 26
D 13

23 A system of equations is shown below:
Equation A: $5x+9y=12$
Equation B: $4x-3y=8$
Which method eliminates one of the variables?

A Multiply equation A by $\frac{1}{3}$ and subtract from B.
B Multiply equation B by 3 and add the result to equation A.
C Multiply equation A by 2 and equation B by -3 and add.
D Multiply equation B by 5 and equation A by 4 and add.

24 If g is a function defined over the set of all real numbers and $g(x-1)=3x^2+5x-7$, then which of the following defines $g(x)$?

A $g(x)=3x^2-x-9$
B $g(x)=3x^2+5x+1$
C $g(x)=3x^2+11x+1$
D $g(x)=3x^2+11x-6$

Rabaa Farrag

25 Which of the following complex numbers is a sum of $\sqrt{-48}$ and $\sqrt{-27}$?

A $-5\sqrt{3}$
B $-7\sqrt{3}$
C $5i\sqrt{3}$
D $7i\sqrt{3}$

26 Given the values of x, $f(x)$ and $g(x)$ in the table below, what is the value of $g(f(2))$?

x	f(x)	g(x)
0	0	-5
1	3	-2
2	6	1
3	9	2
4	12	3
5	15	7
6	18	9

A 2
B 3
C 6
D 9

27 Rita, a junior, is Running for president of the key club. There are 9 other juniors running for the same position. If, historically, a junior has a 1 in 3 chance of being elected president of the club, what are Rita's chances of becoming president of the key club?

A $\frac{1}{3}$
B $\frac{1}{9}$
C $\frac{1}{10}$
D $\frac{1}{30}$

28 What is the value of $f(x) = \frac{x^3}{4}$, when $x = \frac{1}{2}$?

A $\frac{1}{32}$
B $\frac{1}{16}$
C $\frac{1}{12}$
D $\frac{1}{8}$

29 For winning a chess tournament, the captain Ayleen, will receive one-third of what is left of a $500 prize after the team's coach gets $20. What remains after Ayleen gets her share will be split evenly among the other team members. If the team has a total of 5 players, how much will Ayleen's teammate Brenda receive?

A $40
B $80
C $96
D $100

Rabaa Farrag

30 The stem and leaf plot below depicts the daily average temperature (rounded to the nearest degree) for Trees Ville over a two-week period with weekend leafs - W. X. Y, and Z. - absent. If the median daily temperature was $65°$, what could be the mode of the data set?

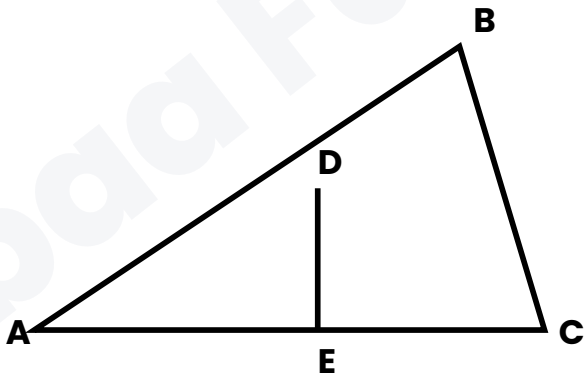
Stem	Leaf
5	8, w, 9
6	X, 2, 4, 4, Y, 6, 6, 9
7	7, 7
8	Z

A $59°$
B $64°$
C $66°$
D $77°$

31 A circle has a circumference of 36π, what is the area?

A 12π
B 18π
C 81π
D 324π

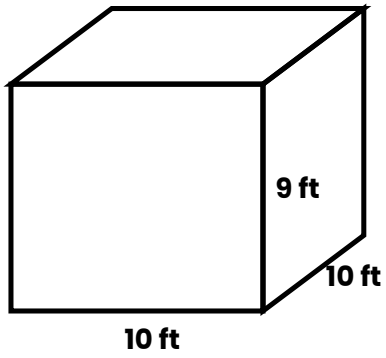
32 In the diagram shown $\overline{DE} \parallel \overline{BC}$, DE measures 4 units, BC measures 6 units, and AC measures 9 units. What is the measurement of EC?



A 3
B 5
C 6
D $\frac{27}{2}$

Rabaa Farrag

33 Rebecca is loading medical supply boxes into a crate. Each supply box is 1.5 feet tall, 1 foot wide, and 2 feet deep. The crate is 9 feet high, 10 feet wide, and 10 feet deep. What is the maximum number of supply boxes can she pack in the crate?



A) 200
B) 300
C) 450
D) 600

34 Let $f(x) = \begin{cases} \sqrt{2x-3} - x, & x > 5 \\ 1 - 2x, & x \leq 5 \end{cases}$. Compute $f(6)$?

A) -11
B) -6
C) -3
D) 3

35 If $f(x) = x^2 - 3x$, what is the value of $f(5)$?

A) 4
B) 8
C) 10
D) 16

36 Triangle JGR is similar to triangle MST. Which statement is NOT always true?

A) $\angle J \cong \angle M$
B) $\angle G \cong \angle T$
C) $\angle R \cong \angle T$
D) $\angle G \cong \angle S$

37 Pedro buys 60 baseball cards to start his collection. Each week, he adds 5 cards to his collection. Lea begins her collection with 80 cards and adds 3 cards to her collection each week. Which of the following equations, when solved, gives the number of weeks, t, after starting their collections that Pedro and Lea will have the same number of cards in their collection?

A) $60 + 3t = 80 + 5t$
B) $60 + 5t = 80 + 3t$
C) $60 - 3t = 80 + 5t$
D) $60 - 5t = 80 + 3t$

Rabaa Farrag

38 Jade has test scores of $82, 75,$ and 78. What does she need to score on her next test so that she has an average test score of 80?

(A) 75
(B) 78
(C) 80
(D) 85

39 What happens to the graph of $y = |x|$ when the equation changes to $y = |x - 2|$?

(A) The graph shifts up 2 units.
(B) The graph shifts left 2 units.
(C) The graph shifts right 2 units.
(D) The graph shifts down 2 units.

40 The frequency table shows the prizes of tickets in a school raffle. There are $1,000$ total ticket for the raffle. A local business purchased 100 tickets, How many free movie tickets should the business expect to win?

Prize	Number of winning tickets
$500	1
$30	9
$10	90
$5	400
Free movie ticket	500

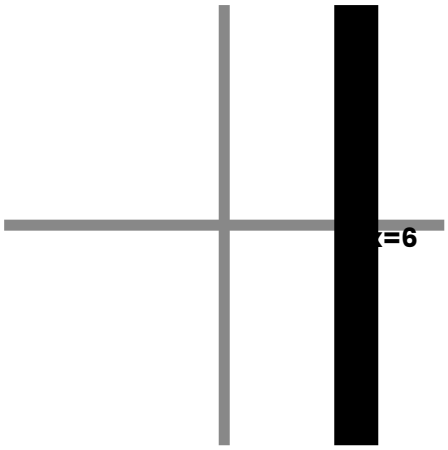
(A) 10 Movie tickets
(B) 20 Movie tickets
(C) 50 Movie tickets
(D) 100 Movie tickets

41 A furnace technician charges $20 per hour and a one-time service charge of $30. Which equation represents the total charge (y) for working (x) hours?

(A) $y = 20x + 30$
(B) $y = 30x + 20$
(C) $y = 20(x - 30)$
(D) $y = 20(x + 30)$

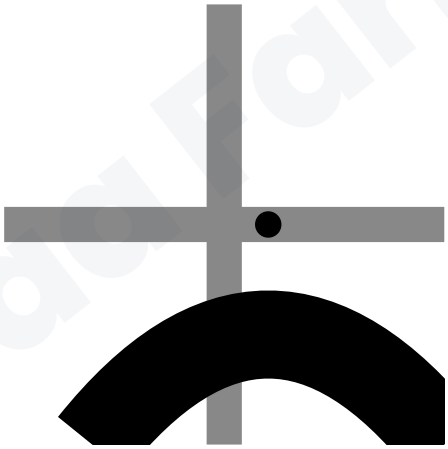
RabaaFarrag

42 What is the equation and slope of the line shown in the grid?



A $y = 6$; slope is $-\frac{1}{6}$
B $x = 6$; slope is zero
C $y = 6$; slope is 6
D $x = 6$; slope is undefined

43 Which graph best represents a quadratic function that has only one zero?



A Parabola crossing x-axis twice
B Parabola whose vertex touches x-axis at one point
C Parabola above x-axis with no x-intercepts
D Inverted parabola crossing x-axis twice

Rabaa Farrag

44 There are 15 plates in a kitchen cabinet. The diameter of each is either 7 inches or 12 inches. The diameter of all 15 plates combined is 140 inches. Which system of equations can be used to find the number of 7-inch plates, x, and the number of 12-inch plates, y, that are in the cabinet?

A $\begin{cases} x + y = 140 \\ 12x + 7y = 15 \end{cases}$

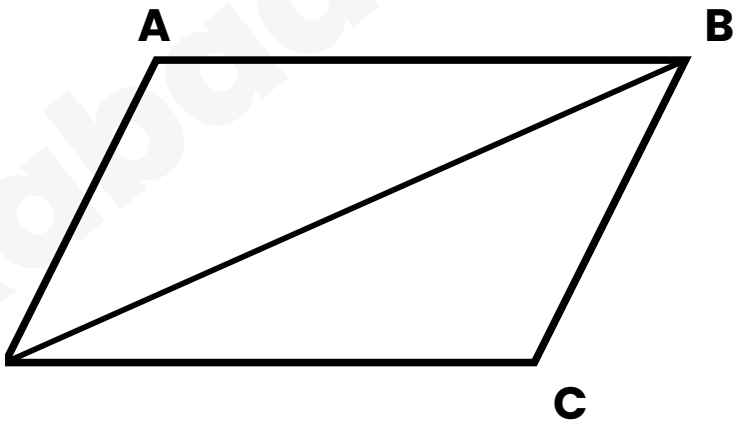
B $\begin{cases} 7x + 12y = 140 \\ 7x + 12y = 15 \end{cases}$

C $\begin{cases} x + y = 15 \\ 7x + 12y = 140 \end{cases}$

D $\begin{cases} x + y = 15 \\ 12x + 7y = 140 \end{cases}$

45 What is the missing reason for the proof?
Given: parallelogram $ABCD$ with diagonal $\overline{BD}$. Proof $\Delta ABD \cong \Delta CDB$

Statement	Reasons
1. AD ≅ BC	1. Definition of parallelogram
2. ∠ADB ≅ ∠CBD	2. Alternate interior angles theorem
3. AB || CD	3. Definition of parallelogram
4. ∠ABD ≅ ∠CDB	4. Alternate interior angles theorem
5. DB ≅ DB	5. Reflexive property of congruence
6. ΔABD ≅ ΔCDB	6. ??



A Reflexive property of congruence

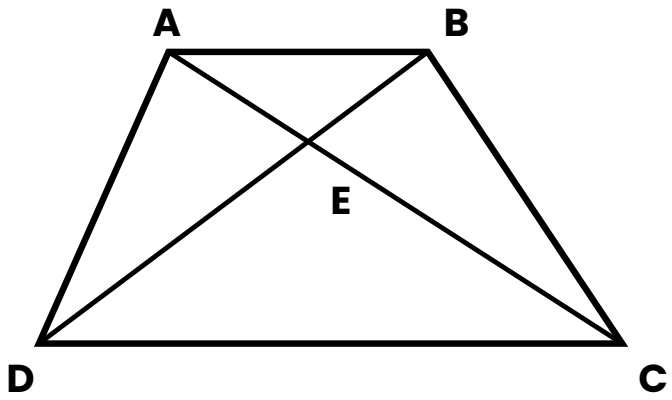
B ASA

C Alternate interior angles theorem

D SSS

Rabaa Farrag

46 In trapezoid $ABCD$ below, $\overline{AB} \parallel \overline{CD}$. If $AE = 5.2$, $AC = 11.7$ and $CD = 10.5$, what is the length of $\overline{AB}$ to the nearest tenth?



(A) 4.7
(B) 6.5
(C) 8.4
(D) 13.1

47 In parallelogram $ABCD$, diagonals $\overline{AC}$ and $\overline{BD}$ intersect at E. Which statement proves $ABCD$ is a rectangle?

(A) $\overline{AC} \cong \overline{BD}$
(B) $\overline{AB} \perp \overline{BD}$
(C) $\overline{AC} \perp \overline{BD}$
(D) $\overline{AC}$ bisects $\angle BCD$

48 Which of the following phrases describes the graph of the line $y = 5$?

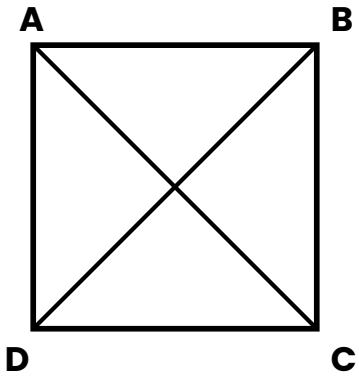
(A) Horizontal line through quadrant I and quadrant II.
(B) Vertical line through quadrant I and quadrant II.
(C) Horizontal line through quadrant I and quadrant IV.
(D) Vertical line through quadrant I and quadrant IV.

49 If the radii of two circles are equal, then the circles are congruent. What is the inverse of the statement above?

(A) If the radii of two circles are equal, then the circles are not congruent.
(B) If two circles are not congruent, then the radii are unequal.
(C) If two circles are congruent, then the radii of the circles are equal.
(D) If the radii of two circles are unequal, then the circles are not congruent.

Rabaa Farrag

50 Given $ABCD$ as marked, what additional information would help you to prove that $ABCD$ is a square?



(A) $\overline{AC} \cong \overline{BD}$
(B) $\overline{AC} \perp \overline{BD}$
(C) $\overline{AC}$ and $\overline{BD}$ bisect each other.
(D) $\overline{AC}$ bisects $\angle DAB$ and $\angle BCD$, $\overline{BD}$ bisects $\angle ABC$ and $\angle CDA$

Rabaa Farrag

TEST #2 • CHECK YOUR WORK

Answer Key

50 ANSWERS

Q	ANSWER	Q	ANSWER
1	B	26	D
2	A	27	D
3	B	28	A
4	C	29	B
5	B	30	C
6	C	31	D
7	C	32	A
8	D	33	B
9	A	34	C
10	C	35	C
11	B	36	B
12	C	37	B
13	D	38	D
14	B	39	C
15	C	40	C
16	B	41	A
17	A	42	D
18	D	43	B
19	C	44	C
20	C	45	B
21	A	46	C
22	D	47	A
23	B	48	A
24	C	49	D
25	D	50	B

Rabaa Farrag

TEST #2 • STEP-BY-STEP

Detailed Solutions

50 SOLUTIONS

1 **Answer: B** $\begin{cases} 3x - y = -27 \\ x + 2y = 16 \end{cases}$

Step 1: Line q passes through $(-9, 0)$ and $(-3, 18)$: slope $= \frac{18-0}{-3+9} = 3$, so $y = 3(x + 9) \Rightarrow 3x - y = -27$.
Step 2: Line v passes through $(-4, 10)$ and $(0, 8)$: slope $= \frac{8-10}{0+4} = -\frac{1}{2}$, y-intercept 8, so $y = -\frac{1}{2}x + 8 \Rightarrow x + 2y = 16$.
Step 3: The system is $3x - y = -27$ and $x + 2y = 16$.

2 **Answer: A** $28°$

Step 1: In right $\triangle ABC$ (right angle at B): $\cos C = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{BC}{AC} = \frac{15}{17}$.
Step 2: $m\angle C = \cos^{-1}\left(\frac{15}{17}\right) \approx 28.1°$.
Step 3: Similar triangles have equal corresponding angles, so $m\angle F = m\angle C \approx 28°$.

3 **Answer: B** $x + 3y = 9$

Step 1: Slope: $m = \frac{6-2}{-9-3} = \frac{4}{-12} = -\frac{1}{3}$.
Step 2: Point-slope form through $(3, 2)$: $y - 2 = -\frac{1}{3}(x - 3)$.
Step 3: Multiply by 3: $3y - 6 = -x + 3 \Rightarrow x + 3y = 9$.

4 **Answer: C** Infinitely many solutions

Step 1: Rewrite the second equation: $10y = -4x + 14 \Rightarrow 4x + 10y = 14$.
Step 2: Divide by 2: $2x + 5y = 7$ — exactly the first equation.
Step 3: The two equations describe the same line, so there are infinitely many solutions.

5 **Answer: B** $4\sqrt{3}$

Step 1: $DC = AC - AD = 12 - 8 = 4$.
Step 2: In a right triangle, the altitude to the hypotenuse gives: (leg)² = (adjacent segment) × (hypotenuse). So $BC^2 = DC \cdot AC = 4 \times 12 = 48$.
Step 3: $BC = \sqrt{48} = 4\sqrt{3}$.

6 **Answer: C** No real numbers

Step 1: Cross-multiply: $5(2x + 3) = 2(5x + 3)$.
Step 2: $10x + 15 = 10x + 6 \Rightarrow 15 = 6$, which is never true.
Step 3: The equation has no solution: no real numbers.

Rabaa Farrag

7 **Answer: C** $(11, 2\sqrt{12})$

Step 1: The radius squared is $(-5)^2 + 12^2 = 25 + 144 = 169$, so the circle is $x^2 + y^2 = 169$.
Step 2: Check each point: $(10, 3) \to 109$; $(-12, 13) \to 313$; $(11, 2\sqrt{12}) \to 121 + 48 = 169$ ✓; $(-8, 5\sqrt{21}) \to 64 + 525 = 589$.
Step 3: Only $(11, 2\sqrt{12})$ lies on the circle.

8 **Answer: D** $x = 115$ and $y = 70$

Step 1: In a cyclic quadrilateral, opposite angles are supplementary.
Step 2: $\angle A + \angle C = 180°$: $80 + (y + 30) = 180 \Rightarrow y = 70$.
Step 3: $\angle B + \angle D = 180°$: $75 + (x - 10) = 180 \Rightarrow x = 115$.

9 **Answer: A** A median

Step 1: A median joins a vertex to the midpoint of the opposite side, splitting that side into two equal bases.
Step 2: The two smaller triangles share the same height from that vertex.
Step 3: Equal bases and equal heights give equal areas — true for every triangle.

10 **Answer: C** $24 + 24\sqrt{3}$

Step 1: The triangle is equilateral. Between the two tangent points on a side, the distance equals the distance between the two circle centers: $2r = 8$.
Step 2: At each corner, the distance from the vertex to the tangent point is $r\sqrt{3} = 4\sqrt{3}$ (a 30-60-90 triangle).
Step 3: Side $= 8 + 2(4\sqrt{3}) = 8 + 8\sqrt{3}$; perimeter $= 3(8 + 8\sqrt{3}) = 24 + 24\sqrt{3}$.

11 **Answer: B** $w = 5$ m, $l = 16$ m

Step 1: Let the width be w, so the length is $3w + 1$.
Step 2: Area: $w(3w + 1) = 80 \Rightarrow 3w^2 + w - 80 = 0 \Rightarrow (3w + 16)(w - 5) = 0$.
Step 3: $w = 5$ m (positive), so $l = 3(5) + 1 = 16$ m.

12 **Answer: C** $\frac{14}{29}$

Step 1: After removing checkers: red $= 16 - 2 = 14$, black $= 16 - 1 = 15$.
Step 2: Total left $= 29$.
Step 3: $P(\text{red}) = \frac{14}{29}$.

13 **Answer: D** 11 weeks

Step 1: After w weeks: Jasmine has $50 + 10w$, her sister has $80 + 7w$.
Step 2: Jasmine has more when $50 + 10w > 80 + 7w \Rightarrow 3w > 30 \Rightarrow w > 10$.
Step 3: The first whole week this happens is week 11.

Rabaa Farrag

14 **Answer: B** $y = 300x + 2000$; slope is 300. This means that for every car Thomas sells, he earns an additional \$300

Step 1: At 0 cars the salary is \$2000, so the y-intercept is 2000.
Step 2: Each extra car adds \$300: slope $= \frac{2300-2000}{1-0} = 300$.
Step 3: Equation $y = 300x + 2000$; the slope means Thomas earns an additional \$300 for every car sold.

15 **Answer: C** $x - y$

Step 1: x is odd and positive; $-y$ is odd and positive.
Step 2: $x - y = x + (-y)$ is the sum of two positive odd numbers, which is always positive and even.
Step 3: (The other choices can be fractions, negative, or odd.)

16 **Answer: B** $3\sqrt{3}$

Step 1: The altitude of an equilateral triangle splits it into two 30-60-90 triangles with hypotenuse 6 and short leg 3.
Step 2: Height $= 3\sqrt{3}$ (or $h = \frac{\sqrt{3}}{2}s = \frac{\sqrt{3}}{2}(6)$).

17 **Answer: A** height: 109 m; distance: 435 m

Step 1: The potato lands when $f(x) = 0$: $x(1 - 0.0023x) = 0 \Rightarrow x = \frac{1}{0.0023} \approx 435$ m.
Step 2: The highest point is at the vertex, halfway: $x \approx 217.4$.
Step 3: $f(217.4) = 217.4 - 0.0023(217.4)^2 \approx 108.7 \approx 109$ m.

18 **Answer: D** SSS

Step 1: $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$ (all sides of a rhombus are equal).
Step 2: $\overline{AC} \cong \overline{AC}$ (shared side).
Step 3: Three pairs of congruent sides: SSS.

19 **Answer: C** $a + 2$

Step 1: In a kite, the axis of symmetry passes through the midpoint of the other diagonal.
Step 2: The horizontal diagonal goes from $x = a$ to $x = a + 4$.
Step 3: Midpoint: $\frac{a+(a+4)}{2} = a + 2$, which is P's x-coordinate.

20 **Answer: C** $13\frac{5}{7}$

Step 1: Vertices: $y = 3x$ meets $y = 0$ at $(0, 0)$; $y = -\frac{1}{2}x + 4$ meets $y = 0$ at $(8, 0)$.
Step 2: $3x = -\frac{1}{2}x + 4 \Rightarrow \frac{7}{2}x = 4 \Rightarrow x = \frac{8}{7}$, $y = \frac{24}{7}$.
Step 3: Area $= \frac{1}{2}(8)\left(\frac{24}{7}\right) = \frac{96}{7} = 13\frac{5}{7}$.

Rabaa Farrag

21 **Answer: A** Circle with center $(-2, 1)$ and radius 2

Step 1: Compare with $(x-h)^2 + (y-k)^2 = r^2$.
Step 2: $(x+2)^2 = (x-(-2))^2$ and $(y-1)^2$: center $(-2, 1)$.
Step 3: $r^2 = 4 \Rightarrow r = 2$.

22 **Answer: D** 13

Step 1: $2(l+w) = 34 \Rightarrow l + w = 17$, and $lw = 60$.
Step 2: Two numbers with sum 17 and product 60: 12 and 5.
Step 3: Diagonal $= \sqrt{12^2 + 5^2} = \sqrt{169} = 13$.

23 **Answer: B** Multiply equation B by 3 and add the result to equation A.

Step 1: The y-terms are $9y$ and $-3y$.
Step 2: Multiply equation B by 3: $12x - 9y = 24$.
Step 3: Adding to A: $(5x + 12x) + (9y - 9y) = 12 + 24$, so y is eliminated.

24 **Answer: C** $g(x) = 3x^2 + 11x + 1$

Step 1: Let $u = x - 1$, so $x = u + 1$.
Step 2: $g(u) = 3(u+1)^2 + 5(u+1) - 7 = 3u^2 + 6u + 3 + 5u + 5 - 7$.
Step 3: $g(u) = 3u^2 + 11u + 1$, so $g(x) = 3x^2 + 11x + 1$.

25 **Answer: D** $7i\sqrt{3}$

Step 1: $\sqrt{-48} = \sqrt{16 \cdot 3} \cdot i = 4i\sqrt{3}$.
Step 2: $\sqrt{-27} = \sqrt{9 \cdot 3} \cdot i = 3i\sqrt{3}$.
Step 3: Sum $= 7i\sqrt{3}$.

26 **Answer: D** 9

Step 1: Work from the inside out: from the table, $f(2) = 6$.
Step 2: Then $g(f(2)) = g(6) = 9$.

27 **Answer: D** $\frac{1}{30}$

Step 1: Rita is one of 10 equally likely junior candidates: $P(\text{Rita, given a junior wins}) = \frac{1}{10}$.
Step 2: $P(\text{a junior wins}) = \frac{1}{3}$.
Step 3: $P(\text{Rita}) = \frac{1}{3} \times \frac{1}{10} = \frac{1}{30}$.

Rabaa Farrag

28 Answer: **A** $\frac{1}{32}$

Step 1: $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$.
Step 2: $f\left(\frac{1}{2}\right) = \frac{1/8}{4} = \frac{1}{32}$.

29 Answer: **B** \$80

Step 1: After the coach: $500 - 20 = 480$.
Step 2: Ayleen gets $\frac{1}{3}(480) = 160$, leaving 320.
Step 3: The other 4 players split it: $\frac{320}{4} = 80$, so Brenda receives \$80.

30 Answer: **C** $66°$

Step 1: There are 14 days, so the median is the average of the 7th and 8th values. With median 65, these are 64 and 66 (the weekend values fill in around them).
Step 2: Known repeats: 64, 66 and 77 each appear twice. If a missing weekend value Y is 6, then 66 appears three times and becomes the mode.
Step 3: 59, 64 and 77 cannot be made the unique mode while keeping the median at 65, so the mode could be $66°$.

31 Answer: **D** 324π

Step 1: $2\pi r = 36\pi \Rightarrow r = 18$.
Step 2: $A = \pi r^2 = 324\pi$.

32 Answer: **A** 3

Step 1: $\overline{DE} \parallel \overline{BC}$, so $\triangle ADE \sim \triangle ABC$.
Step 2: $\frac{AE}{AC} = \frac{DE}{BC} = \frac{4}{6} = \frac{2}{3} \Rightarrow AE = \frac{2}{3}(9) = 6$.
Step 3: $EC = 9 - 6 = 3$.

33 Answer: **B** 300

Step 1: Fit the boxes along each dimension: height $\frac{9}{1.5} = 6$, width $\frac{10}{1} = 10$, depth $\frac{10}{2} = 5$.
Step 2: Total $= 6 \times 10 \times 5 = 300$ boxes (same as volume $900 \div 3 = 300$).

34 Answer: **C** -3

Step 1: $x = 6 > 5$, so use the first rule: $f(6) = \sqrt{2(6) - 3} - 6$.
Step 2: $= \sqrt{9} - 6 = 3 - 6 = -3$.

Rabaa Farrag

35 **Answer: C** 10

Step 1: $f(5) = 5^2 - 3(5)$.
Step 2: $= 25 - 15 = 10$.

36 **Answer: B** $\angle G \cong \angle T$

Step 1: The order of letters gives the correspondence: $J \leftrightarrow M$, $G \leftrightarrow S$, $R \leftrightarrow T$.
Step 2: So $\angle J \cong \angle M$, $\angle G \cong \angle S$, $\angle R \cong \angle T$ are always true.
Step 3: $\angle G \cong \angle T$ is not a corresponding pair, so it is not always true.

37 **Answer: B** $60 + 5t = 80 + 3t$

Step 1: Pedro: starts at 60 and adds 5 a week: $60 + 5t$.
Step 2: Lea: starts at 80 and adds 3 a week: $80 + 3t$.
Step 3: Equal collections: $60 + 5t = 80 + 3t$.

38 **Answer: D** 85

Step 1: An average of 80 over 4 tests needs a total of 320.
Step 2: Current total: $82 + 75 + 78 = 235$.
Step 3: Needed score: $320 - 235 = 85$.

39 **Answer: C** The graph shifts right 2 units.

Step 1: Replacing x with $x - h$ shifts a graph h units right.
Step 2: $y = |x - 2|$ is $y = |x|$ shifted right 2 units (its vertex moves from $(0, 0)$ to $(2, 0)$).

40 **Answer: C** 50 Movie tickets

Step 1: 500 of the 1000 tickets win a free movie ticket: probability $= \frac{500}{1000} = \frac{1}{2}$.
Step 2: Expected number with 100 tickets: $100 \times \frac{1}{2} = 50$ movie tickets.

41 **Answer: A** $y = 20x + 30$

Step 1: The hourly part is $20x$; the one-time charge is 30.
Step 2: $y = 20x + 30$.

42 **Answer: D** $x = 6$; slope is undefined

Step 1: The line is vertical, crossing the x-axis at 6: every point has $x = 6$.
Step 2: Equation $x = 6$; a vertical line has undefined slope.

Rabaa Farrag

43 **Answer: B** Parabola whose vertex touches x-axis at one point

Step 1: The zeros of a quadratic are where its graph meets the x-axis.
Step 2: Exactly one zero means the parabola just touches the x-axis at its vertex.

44 **Answer: C** $\begin{cases} x + y = 15 \\ 7x + 12y = 140 \end{cases}$

Step 1: Number of plates: $x + y = 15$.
Step 2: Sum of diameters: $7x + 12y = 140$.

45 **Answer: B** ASA

Step 1: $\angle ABD \cong \angle CDB$ and $\angle ADB \cong \angle CBD$ (alternate interior angles of parallel sides).
Step 2: $\overline{BD} \cong \overline{DB}$ is the side between those angles.
Step 3: Angle–Side–Angle: ASA.

46 **Answer: C** 8.4

Step 1: $EC = AC - AE = 11.7 - 5.2 = 6.5$.
Step 2: $\overline{AB} \parallel \overline{CD}$ gives $\triangle ABE \sim \triangle CDE$: $\frac{AB}{CD} = \frac{AE}{EC} = \frac{5.2}{6.5} = 0.8$.
Step 3: $AB = 0.8 \times 10.5 = 8.4$.

47 **Answer: A** $\overline{AC} \cong \overline{BD}$

Step 1: A parallelogram is a rectangle exactly when its diagonals are congruent.
Step 2: So $\overline{AC} \cong \overline{BD}$ proves $ABCD$ is a rectangle. (Perpendicular diagonals or a diagonal bisecting angles would make it a rhombus.)

48 **Answer: A** Horizontal line through quadrant I and quadrant II.

Step 1: $y = 5$ is a horizontal line 5 units above the x-axis.
Step 2: It contains points with positive x (quadrant I) and negative x (quadrant II).

49 **Answer: D** If the radii of two circles are unequal, then the circles are not congruent.

Step 1: Statement: if p (radii equal), then q (circles congruent).
Step 2: The inverse is "if not p, then not q".
Step 3: "If the radii of two circles are unequal, then the circles are not congruent."

Rabaa Farrag

50 **Answer: B** $\overline{AC} \perp \overline{BD}$

Step 1: The markings show $ABCD$ is a rectangle (or has equal sides); a square needs both.
Step 2: A rectangle whose diagonals are perpendicular is also a rhombus, hence a square.
Step 3: So $\overline{AC} \perp \overline{BD}$ is the needed information.

Rabaa Farrag